Technological Ecosystem Simulation

Exercise — Part 1: Original text / Part 2: English version

Part 1 — Original text (as provided)

```
EXERCISE: TECHNOLOGICAL ECOSYSTEM SIMULATION

Develop a program that simulates the interaction over time
between three entities:
  - Robots (R1)
  - Drones (D2)
  - Energy Stations (E)

The simulation progresses day by day. The program stops and the simulation
ends immediately if the number of Robots, Drones or Stations reaches 0.

SYSTEM RULES (Apply in this exact order if the days coincide):

1. EVERY DAY (Day 1, 2, 3...):
   - The existing Energy Stations (E) are duplicated.

2. EVERY 2 DAYS (Day 2, 4, 6...):
   - Each Robot (R1) needs to consume 4 Stations (E) to recharge.
   - If the system has enough energy for it, it consumes the 4 and it doubles.
   - If there is not enough energy in the system, the Robot turns off (dies).

3. EVERY 3 DAYS (Day 3, 6, 9...):
   - Phase 1 (Attack): For each complete group of 2 Drones, 1 Robot is destroyed.
   - Phase 2 (Recharge): Each Drone (D2) needs to consume 3 Stations (E).
     If there is enough energy, it consumes it and it doubles.
     If there is not enough energy, the Drone turns off (dies).

4. EVERY 4 DAYS (Day 4, 8, 12...):
   - Phase 1 (Overload): For every 50 Stations (E) accumulated, an overload
     occurs in the system that destroys 2 Robots.
   - Phase 2 (Hacking): For each complete group of 4 Robots (R1) alive,
     they manage to hack and destroy 1 Drone.

INITIAL CONDITIONS:
With these values, the simulation should survive at least 25 days without
any population dying out.

  * Robots (R1): 10
  * Drones (D2): 6
  * Estaciones (E): 50
```

Part 2 — English version

Exercise: Technological Ecosystem Simulation

Write a program that simulates how three kinds of entities interact over time:

- **Robots** (R1)
- **Drones** (D2)
- **Energy Stations** (E)

The simulation advances **one day at a time**. The program must stop, and the simulation ends immediately, as soon as the number of Robots, Drones or Energy Stations reaches 0.

System rules

When several rules fall on the same day, apply them in the order listed below.

Rule 1 — Every day (days 1, 2, 3, ...)

- Energy Stations (E) double: the existing stations are duplicated.

Rule 2 — Every 2 days (days 2, 4, 6, ...)

- Each Robot (R1) must consume 4 Energy Stations (E) to recharge.
  - If the system has enough energy, the Robot consumes the 4 stations and doubles.
  - If there is not enough energy, the Robot shuts down (dies).

Rule 3 — Every 3 days (days 3, 6, 9, ...)

- **Phase 1 – Attack:** for every complete group of 2 Drones, 1 Robot is destroyed.
- **Phase 2 – Recharge:** each Drone (D2) must consume 3 Energy Stations (E).
  - If there is enough energy, the Drone consumes it and doubles.
  - If there is not enough energy, the Drone shuts down (dies).

Rule 4 — Every 4 days (days 4, 8, 12, ...)

- **Phase 1 – Overload:** for every 50 Energy Stations (E) accumulated, an overload occurs in the system and destroys 2 Robots.
- **Phase 2 – Hacking:** for every complete group of 4 surviving Robots (R1), they hack and destroy 1 Drone.

Initial conditions

With these starting values, the simulation should survive at least 25 days without any population dying out.

Entity	Symbol	Initial amount
Robots	R1	10
Drones	D2	6
Energy Stations	E	50